PyGameMaker — Tutorial 16: Fruit Basket — Teacher Guide

Companion for the in-app tutorial (**Help > Tutorials > Fruit Basket: Catch and Avoid!**, 6 pages) and the matching student handout and worksheet. No earlier tutorial is required — this is the right first project for a student who isn't ready for Tutorial 15's "remember what you're holding" variable yet. Every action in this game is a single, ready-made block; nothing is created, named, or read back by the student.

Overview

Students build a complete catch/avoid/win game without ever touching the idea of a variable. A basket moves along the bottom of the room; fruit and bombs are placed once and never move or respawn, so there's nothing to recycle and nothing to remember between frames. Four phases: a moving basket; placing fruit and bombs with no behaviour yet; wiring up collisions (catch a fruit for points, touch a bomb and the room restarts); and a win check once every fruit is gone.

> If a student asks "how does the game know I caught every fruit", the honest answer is: it counts, every single step, how many obj_fruit instances still exist in the room — the **If count of** block. Nothing is stored or remembered on purpose; the count is just read fresh each time.

Suggested Timing (30-35 minutes)

The tutorial says 10-15 minutes for a confident student; allow extra time for the first-ever collision event and the first-ever room restart, since both are new ideas even though neither needs a variable.

Segment	Time	What happens
Recap and predict	5 min	Show the finished game; ask what happens if you touch a bomb
Phase 1: moving basket	5 min	Page 2
Phase 2: fruit and bombs	5 min	Page 3 (placement only, no behaviour yet)
Phase 3: catching and avoiding	10 min	Page 4 (two new collision events)
Phase 4: winning	5-10 min	Page 5
Worksheet / exit ticket	5 min	Worksheet Parts A-C

Step-by-Step Walkthrough and Common Issues

Phase 1: moving basket

- Three keyboard events, exactly as in Tutorial 2 / Tutorial 9 / Tutorial 15. The basket may leave the screen; that is expected and not fixed in this tutorial.

Phase 2: fruit and bombs

- The most common placement mistake: a fruit placed too close to where the basket starts, so the basket "catches" it just by existing, before the player presses a key. Walk students through checking the basket's starting box against each fruit's position if this happens.

- Nothing falls and nothing moves here — a student coming from Tutorial 9 or Tutorial 15 may expect a **Create** event setting a fall speed. There isn't one; that's the point.

Phase 3: catching and avoiding

- obj_fruit's collision handler lives on obj_player, not on the fruit itself — same **Destroy other instance** pattern as every other catch-game tutorial (Tutorial 2, 9, 15). A student who writes *Destroy this instance* destroys the basket instead.
- **Restart Room** is new vocabulary this tutorial introduces. It is deliberately NOT a punishment with lives or a game-over screen — make sure students notice it just starts the current room over, fruit and all, so touching a bomb costs a little time, not the game.
- A student who skips the **Draw** event will still have a working game — the score just isn't visible. This is an easy, low-stakes thing to notice and fix without re-explaining anything.

Phase 4: winning

- The win check is a **Step** event on obj_player, exactly Tutorial 9's pattern (If count of obj_coin becomes If count of obj_fruit) — a student who has done Tutorial 9 will recognise this immediately.
- If a student placed zero fruit in Phase 2 (unlikely, but it happens), the win room appears the instant the game starts. This is a good moment to point out that the count really is checked fresh every step, not stored.

> **Reference projects.** Download the checkpoint projects from the wiki (16_fruit_basket_checkpoints.zip): one project for the end of each phase. Give a stuck student the previous phase rather than troubleshooting their collisions from scratch.

Vocabulary Introduced

Term	What it means here
Collision	An event that fires the moment two objects touch
Instance count	How many copies of an object exist right now
Restart Room	An action that puts the current room back exactly how it started
Step event	An event that runs every single frame of the game, over and over

Discussion Questions

- Why doesn't obj_fruit need any events of its own, when obj_coin in Tutorial 9 needed a Create event and an Outside Room event?
- What's the difference between **Restart Room** and **Restart Game**? When would you want one instead of the other?
- Why can the win check in Phase 4 "just count" instead of needing to remember anything from a previous step?
- What other games let you "just try again" instead of punishing a mistake with a game-over screen?

Differentiation

- **Support:** give the Phase 2 checkpoint, so a struggling student only has to wire up the two collision events in Phase 3-4.

- **Extension:** a second fruit type worth more points; a timer that shows how fast the room was cleared; a basket that can also move up and down to reach a second row of fruit.

Worksheet Answer Key

Part A: 1-C, 2-B, 3-A, 4-D.

Part B: 1. obj_player, Collision with obj_fruit. 2. obj_player, Collision with obj_bomb. 3. obj_player, Step. 4. obj_win_text, When drawing. 5. obj_win_text, Key press: space.

Part C:

1. obj_fruit never moves and never changes — it just sits in the room waiting to be touched. All the *behaviour* ("what happens when you touch me") lives on obj_player's collision events instead, the same way obj_coin in Tutorial 9 only needed a fall speed and obj_fruit here needs nothing at all, since it doesn't even fall.
2. The game would check how many bombs are left instead of how much fruit is left — since bombs are never destroyed by anything in this game, that count would never reach zero, so the win room would never appear.
3. If a bomb sat close to a fruit, a student aiming for the fruit could touch the bomb by accident while lining up the catch, turning "collect everything" into a precision-dodging challenge instead of the intended "move toward what you want" game.

Parts D and E: completion and reflection.

Rubric: Fruit Basket

Level	What the game shows
4 - Complete	The basket moves; fruit and bombs sit in the room; catching a fruit adds 10 to the score and removes it; touching a bomb restarts the room; collecting every fruit shows YOU WIN! and SPACE restarts
3 - Working	Moving and catching fruit work; the bomb restart or the win screen is missing or incorrect
2 - Partly there	The basket moves and fruit/bombs are placed, but catching or avoiding doesn't work correctly
1 - Started	The basket exists and moves; nothing else is placed yet

End-of-Session Checklist

- [ ] Every student has a saved project (Ctrl+S)
- [ ] Every student reached at least Phase 3 (catching a fruit works end to end)
- [ ] Note who needs the checkpoint project next time